

Prescribed finite modality in a conserved-reactivity encounter process

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 (Dated: Internal theory-section build harness, July 2026)

I. ANALYTICAL CONSTRUCTION

A. Prescribed finite modality from separated exposure clocks

The conserved budget does more than perturb the position of a pre-existing mode. It admits encounter processes with any prescribed *fixed finite* number of reaction-time modes. We state this result in an exact quotient of a two-particle Doi process; the construction depends on the prescribed dimension and mode count.

Fix finite $d \geq 2$, finite $m \geq 1$, a transverse period $W > 0$, a contact radius $0 < a < W/2$, positive scales $\ell_0, \rho > 0$, and a compact interval $I = [\tau, T] \subset (0, \infty)$. On $\mathbb{R} \times \mathbb{R} \times \mathbb{T}_W^{d-1}$, consider the independent midpoint and relative processes

$$\begin{aligned} dZ_t &= -\gamma(Z_t - \bar{z}) dt + \varepsilon \sqrt{D_0} dW_t, \\ dR_{\parallel,t} &= -\gamma R_{\parallel,t} dt + 2\varepsilon \sqrt{D_0} dW_{\parallel,t}, \\ dR_{\perp,t} &= 2\varepsilon \sqrt{D_0} dW_{\perp,t} \pmod{W}. \end{aligned} \quad (1)$$

The mutually independent initial laws satisfy

$$\begin{aligned} Z_0 &\sim N(z_0, \varepsilon^2 D_0 / (2\gamma)), & D_0, \gamma, u_0 &> 0, & z_0 &\neq \bar{z}, \\ R_{\parallel,0} &\sim N(r_{\parallel,0}, \varepsilon^2 u_0^2), & u_0^2 &< 4D_0/\gamma, & \Sigma_{\perp,0} &\succ 0, \\ R_{\perp,0} &\sim \text{WN}(r_{\perp,0}, \varepsilon^2 \Sigma_{\perp,0}), \end{aligned} \quad (2)$$

All driving Brownian motions are also independent. Thus $\mu(t) = \bar{z} + (z_0 - \bar{z})e^{-\gamma t}$ is strictly monotone and $\text{Var } Z_t = \varepsilon^2 D_0 / (2\gamma)$. Under the minimum-image convention, assume the deterministic relative trajectory $r_*(t) = (r_{\parallel,0}e^{-\gamma t}, r_{\perp,0})$ obeys $\sup_{t \in I} |r_*(t)|_{\text{mi}} \leq a - \eta$ for some $\eta > 0$.

Choose target times

$$\begin{aligned} \tau < t_1 < \dots < t_m < T, & \quad c_j = x(t_j), \\ x(t) &= \frac{\text{sgn}(\mu')\mu(t)}{\ell_0}, \end{aligned} \quad (3)$$

so that $x'(t)$ is bounded away from zero. At the corresponding physical centres, place normalized Gaussian catalyst slabs of width $\varepsilon\rho$. Their nonnegative weights belong to a compact subset of the simplex,

$$\mathcal{W}_{w_*} = \left\{ \mathbf{w} : \sum_{j=1}^m w_j = 1, w_j \geq w_* > 0 \right\}, \quad 0 < w_* \leq \frac{1}{m}, \quad (4)$$

and define

$$\begin{aligned} \phi_{j,\varepsilon}(z) &= \frac{\exp[-(z - \mu(t_j))^2 / (2\varepsilon^2 \rho^2)]}{\sqrt{2\pi} \varepsilon \rho}, \\ K_{B,\mathbf{w},\varepsilon}(Z, R) &= \mathbf{1}_{\{|R|_{\text{mi}} < a\}} \frac{B}{W^{d-1}} \sum_{j=1}^m w_j \phi_{j,\varepsilon}(Z). \end{aligned} \quad (5)$$

The slab normalization makes B the same installed centre-space budget for every allocation. Set $K_{B,\mathbf{w},\varepsilon} = BV_{\mathbf{w},\varepsilon}$, and let $T_0(t)$ and q_0 denote the unkilld semi-group and initial law. Then

$$\begin{aligned} G_{\varepsilon,\mathbf{w}}(t) &= B^{-1} \mathbb{E}_0[K_{B,\mathbf{w},\varepsilon}(Z_t, R_t)] = \langle V_{\mathbf{w},\varepsilon}, T_0(t)q_0 \rangle, \\ f_{B,\varepsilon,\mathbf{w}}(t) &= \mathbb{E}_0 \left[K_{B,\mathbf{w},\varepsilon}(Z_t, R_t) e^{-\int_0^t K_{B,\mathbf{w},\varepsilon}(Z_s, R_s) ds} \right]. \end{aligned} \quad (6)$$

Theorem 1 (exact prescribed finite modality).— Fix finite $d \geq 2$, finite $m \geq 1$, the data above, and the compact window I . There is $\varepsilon_0 > 0$ such that, for every $0 < \varepsilon < \varepsilon_0$ and every $\mathbf{w} \in \mathcal{W}_{w_*}$, the unit-budget free-exposure clock has exactly m nondegenerate maxima and $m - 1$ nondegenerate minima on I , ordered alternately, and no stationary endpoint. For each fixed such ε , there is $B_0(\varepsilon) > 0$, uniform over $\mathcal{W}_{w_*}$, such that the budget-rescaled Doi reaction-time density $f_{B,\varepsilon,\mathbf{w}}/B$ has the same complete stationary signature for every $0 < B < B_0(\varepsilon)$. Multiplication by B leaves the stationary points unchanged, so the conclusion also holds for the physical density $f_{B,\varepsilon,\mathbf{w}}$.

The mechanism can be seen without invoking a normal form. Gaussian convolution reduces the longitudinal part of the unit-budget exposure to

$$\begin{aligned} H_{\sigma,\mathbf{w}}(x) &= \sum_{j=1}^m w_j \exp \left[-\frac{(x - c_j)^2}{2\sigma^2} \right], \\ \sigma &= \frac{\varepsilon}{\ell_0} \sqrt{\frac{D_0}{2\gamma} + \rho^2}. \end{aligned} \quad (7)$$

Writing $L = \partial_x \log H$ and letting $\bar{c}(x)$ and $\text{Var}_{\pi(x)}(c)$ denote the posterior mean and variance of the centres gives the exact identities

$$L(x) = \frac{\bar{c}(x) - x}{\sigma^2}, \quad L'(x) = \frac{\text{Var}_{\pi(x)}(c)}{\sigma^4} - \frac{1}{\sigma^2}. \quad (8)$$

For the adjacent pair $c_j < c_{j+1}$, the weighted crossover is

$$s_j = \frac{c_j + c_{j+1}}{2} + \frac{\sigma^2}{c_{j+1} - c_j} \log \frac{w_j}{w_{j+1}}. \quad (9)$$

Writing $q_k(x) = w_k \exp[-(x - c_k)^2 / (2\sigma^2)]$, the adjacent ratio obeys $q_{j+1}/q_j = 1/9$ at $s_j - \sigma^2 \log(9)/(c_{j+1} - c_j)$ and $q_{j+1}/q_j = 9$ at the corresponding plus edge. Thus both adjacent components remain uniformly present throughout the crossover, while all nonadjacent components are exponentially small. Equation (8) then yields one simple minimum at

$$r_j = s_j + O(\sigma^4), \quad L'(r_j) = \frac{(c_{j+1} - c_j)^2}{4\sigma^4} + O(\sigma^{-2}). \quad (10)$$

Each centre similarly contributes one simple maximum at $p_j = c_j + O(\exp[-q/\sigma^2])$, with $L'(p_j) = -\sigma^{-2} + o(\sigma^{-2})$. These constructions give $2m - 1$ simple stationary points. They are exhaustive: after multiplication by a positive Gaussian factor, H' is an exponential polynomial

$$\sum_{j=1}^m (a_j + b_j x) e^{\lambda_j x}, \quad \lambda_1 < \dots < \lambda_m, \quad (11)$$

which has at most $2m - 1$ real zeros counted with mul-

tiplicity. The extreme affine-exponential terms dominate in the two tails, so all zeros lie in a compact interval and real analyticity makes their number finite. A generalized Rolle induction then proves the bound by differentiating twice to remove the lowest-exponent affine term.

The remaining relative-coordinate contact probability is a positive slow factor. Uniform bounds on its first two logarithmic time derivatives, peak boxes of width $O(\sigma^2)$, valley boxes of width $O(\sigma^4)$, and a posterior-sector sign certificate exclude stationary points on the full box complement. Consequently the slow factor moves maxima by $O(\sigma^2)$ and minima by $O(\sigma^4)$ without changing their number or type. Finally, on every fixed positive ε , compact-positive-time mixed-jet convergence

$$\sup_{\mathbf{w} \in \mathcal{W}_{w_*}} \left\| \frac{f_{B,\varepsilon,\mathbf{w}}}{B} - G_{\varepsilon,\mathbf{w}} \right\|_{C^2(I)} \rightarrow 0 \quad (B \downarrow 0) \quad (12)$$

preserves the disjoint root tubes, signed curvature margins, complement margin, and endpoint slopes.

The quantifier order is essential: the geometry and admissible scales may depend on fixed (d, m) ; one first fixes a sufficiently small $\varepsilon > 0$ and only then chooses $B < B_0(\varepsilon)$. The theorem does not provide a useful uniform lower bound for B_0 , a uniform-in- d or uniform-in- m construction, an event-mass floor, or a stationary-point count outside I ; it also uses a whole-window contact-interior condition whose small-noise limit is asymptotically saturated contact. Nontrivial contact, observability at a common positive budget, and finite-parameter robustness remain separate numerical validation gates.